Assignment 5 - Agentic scientific discovery and research results

Submission Date: Sunday, November 8th, 2026, 11:59pm

Class presentations: Monday, November 9th, 2026

In this assignment you should build on the refined method and preliminary research results developed in Assignment 4. Find a meaningful way to apply agentic scientific discovery methods to your GeoAI coursework project topic. At this stage you should try fully run the associated analysis and illustrate what you have found to the class verbally. The aim is to move your project one step beyond the previous pilot and demonstrate, in particular, how an agent can help you investigate a scientific question.

Please make an effort to **use these agentic methods for scientific discovery**, rather than purely as a productivity tool (which is being deprioritized moving forward in this class).

Create a 5-minute presentation to be presented on Monday, November 9th. Briefly introduce your problem and research question (to remind the class), and then explain the agentic discovery method you applied, and focus on the new results and what they mean for your coursework project. Truth data will be an important element of most projects. If you are taking a pure simulation approach, then consider how secondary datasets could help to verify and validate your approach.

Based on what we covered in class this week, an agentic discovery workflow should use results to guide further investigation. For example, after an initial set of baseline methods and results, an agent might propose a set of improvements to run, again inspect the results against the baseline, and then cyclically use AI reasoning to continue to make more improvements. You should execute at least one such cycle, including a follow-up test informed by the earlier result (but more is better, and you should report the cycles to the class). You can supervise the workflow and approve each step (e.g., you are the human-in-the-loop).

Simply using AI to write code or reformat the results from Assignment 4 is insufficient. State explicitly what is new and why it helps answer your research question.

Table 1 Research progress aspects to address.

Component	Question to address
Research question and progress	Recap on what question you are pursuing, and what have you accomplished since Assignment 4 (including refinements)? Identify the hypothesis, uncertainty or methodological choice investigated in this assignment.
Agentic discovery method	What did the agent propose, execute, evaluate and revise? Include a workflow diagram showing the feedback loop, the tools used, and where you made decisions. Explain how an observed result informed the follow-up test.
Experiments and comparison	Show an executed baseline or comparator, an initial agent experiment and a follow-up test. Your previous pilot may serve as the baseline if the comparison remains fair. State the data, sample size, evaluation criteria and relevant settings. Keep evaluation conditions comparable, or explain differences that limit the comparison.
Results and graphics	Present 1-2 substantive new result graphics, such as a map, comparison plot, error analysis or sensitivity plot. Explain what was measured, what changed relative to the baseline, and what the evidence supports. Include unsuccessful experiments where they help explain what you learned.
Validation and human review	Explain how you checked/verified the data/code and outputs. Explain how you stopped the AI cheating. Give specific examples of agent suggestions you accepted, changed, rejected, etc., and why.

Interpretation and next steps	What have you learned about your scientific question and the usefulness of the agentic approach? State the main limitations, and the next step toward the final paper.

A small, well-checked investigation is sufficient. If you use synthetic data, label it clearly and explain what still needs to be tested with real observations. Separate measured findings from final expected outcomes.

Please submit your presentation online as a PDF by 11:59pm on Sunday, November 8th. You will present it to the class on Monday, November 9th. The submitted slides must already include the executed discovery workflow and new results. You may improve the graphics and extend the analysis before presenting, but clearly identify any results added or changed since submission. Prepare professional, legible slides with consistent formatting, slide numbers, source acknowledgments and an AI disclaimer identifying the tools used, their contributions and your own checks. Use supporting slides for full references, experimental details and a concise record of the agent's proposals, runs, results and your decisions. Retain the associated code and outputs so you can explain how the findings were produced. The main presentation should stand on its own and keep to five minutes (not go over, and do not ramble without a clear planned narrative).

Treat this as a professional research briefing. Give each graphic a clear takeaway, readable labels, units, legends and data sources as appropriate. Briefly establish the context, then spend most of your time explaining the discovery process, the evidence and what you learned. Be prepared to discuss both the scientific findings and the practical value and limitations of the agentic workflow with our guest.

Table 2 Assessment criteria.

Criterion	Evidence	Weight
Agentic discovery method and progress	Clear connection to the coursework research question. Meaningful progress beyond Assignment 4. An executed feedback loop in which an earlier result informs a follow-up test, with explicit agent and human roles.	30%
Research evidence and interpretation	Informative new results and graphics, a fair baseline comparison, appropriate validation, and a defensible interpretation of findings and limitations.	30%
Critical human review	Specific decisions about agent proposals and outputs, checks of data and code, an honest experimental record, and justified next steps for the project.	20%
Presentation quality and delivery	Professional graphics and slides. Proper citations and a suitable AI usage statement. A clear, well-rehearsed oral explanation that keeps to five minutes.	20%

Students remain responsible for the accuracy and defensibility of all submitted material. Verify citations and any AI-generated code, and explain the scientific reasoning in your own words. An agent's explanation is a hypothesis to investigate, not evidence by itself. Agreement between agents is not independent scientific validation.